

Fall 1992

Problem 1 (Fa80, Fa92). Are the matrices give below similar ?

$$A = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 1 \\ -1 & 0 & 2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Problem 2 (Fa92). Let $\mathfrak{I}$ denote the ideal in $\mathbb{Z}[x]$, the ring of polynomials with coefficients in $\mathbb{Z}$, generated by $x^3 + x + 1$ and 5. Is $\mathfrak{I}$ a prime ideal?

Problem 3 (Fa92, Sp09). 1. How many zeros does the polynomial defined by $p(z) = 2z^5 + 4z^2 + 1$ have in the disc $|z| < 1$?

2. How many zeros does the same polynomial have on the real axis?

Problem 4 (Fa79, Su81, Fa92). Let $y = y(x)$ be a solution of the differential equation $y'' = -|y|$ with $-\infty < x < \infty$, $y(0) = 1$ and $y'(0) = 0$.

1. Show that y is an even function.

2. Show that y has exactly one zero on the positive real axis.

Problem 5 (Fa92). Let G be a group and H and K subgroups such that H has a finite index in G . Prove that $K \cap H$ has a finite index in K .

Problem 6 (Fa92). Let (X_1, d_1) and (X_2, d_2) be metric spaces and $f : X_1 \rightarrow X_2$ a continuous surjective map such that $d_1(p, q) \leq d_2(f(p), f(q))$ for every pair of points p, q in X_1 .

1. If X_1 is complete, must X_2 be complete? Give a proof or a counterexample.

2. If X_2 is complete, must X_1 be complete? Give a proof or a counterexample.

Problem 7 (Fa92, Sp13). Evaluate $\int_C \frac{e^z}{z(2z+1)^2} dz$ where C is the unit circle with counterclockwise orientation.

Problem 8 (Fa92). Let $\mathbb{F}$ be a field, V a finite dimensional vector space over $\mathbb{F}$, and T a linear transformation of V into V whose minimal polynomial, μ , is irreducible over $\mathbb{F}$.

1. Let v be a nonzero vector in V and let V_1 be the subspace spanned by v and its images under the positive powers of T . Prove that $\dim V_1 = \deg \mu$.
2. Prove that $\deg \mu$ divides $\dim V$.

Problem 9 (Fa92). Let the function f be analytic in the region $|z| > 1$ of the complex plane. Prove that if f is real valued on the interval $(1, \infty)$ of the real axis, then f is also real valued on the interval $(-\infty, -1)$.

Problem 10 (Fa92). How many Sylow 2-subgroups does the dihedral group D_{2n} of order $2n$ have, when n is odd?

Problem 11 (Su80, Fa92). Let E be a finite dimensional vector space over a field $\mathbb{F}$. Suppose $B : E \times E \rightarrow \mathbb{F}$ is a bilinear map, not necessarily symmetric. Define subspaces

$$E_1 = \{x \in E \mid B(x, y) = 0 \text{ for all } y \in E\}$$

$$E_2 = \{y \in E \mid B(x, y) = 0 \text{ for all } x \in E\}$$

Prove that $\dim E_1 = \dim E_2$.

Problem 12 (Fa92). Let $\{f_n\}$ be a sequence of real valued C^1 functions on $[0, 1]$ such that, for all n ,

$$|f'_n(x)| \leq \frac{1}{\sqrt{x}} \quad \text{for } 0 < x \leq 1$$

$$\int_0^1 f_n(x) \, dx = 0$$

Prove that the sequence has a subsequence that converges uniformly on $[0, 1]$.

Problem 13 (Sp90, Fa92). Let a be a positive real number. Evaluate the improper integral

$$\int_0^\infty \frac{\sin x}{x(x^2 + a^2)} \, dx$$

Problem 14 (Fa92). Let f and g be positive continuous functions on $\mathbb{R}$, with $g \leq f$ everywhere. Assume the initial value problem

$$\frac{dx}{dt} = f(x) \quad x(0) = 0$$

has a solution defined on all of $\mathbb{R}$. Prove that the initial value problem

$$\frac{dx}{dt} = g(x) \quad x(0) = 0$$

also has a solution defined on all of $\mathbb{R}$.

Problem 15 (Fa82, Fa92). Let

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \mid a, b \in \mathbb{R}, a > 0 \right\} \quad \text{and} \quad N = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \mid b \in \mathbb{R} \right\}$$

1. Show that N is a normal subgroup of G and prove that G/N is isomorphic to $\mathbb{R}$.
2. Find a normal subgroup N' of G satisfying $N \subset N' \subset G$ (where the inclusions are proper), or prove that there is no such subgroup.

Problem 16 (Sp80, Fa92). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuously differentiable. Assume the Jacobian matrix $(\partial f_i / \partial x_j)$ has rank n everywhere. Suppose f is *proper*, that is, $f^{-1}(K)$ is compact whenever K is compact. Prove $f(\mathbb{R}^n) = \mathbb{R}^n$.

Problem 17 (Fa92). Let s be a real number, and let the function u be defined in $\mathbb{C} \setminus (-\infty, 0]$ by

$$u(re^{i\theta}) = r^s \cos s\theta \quad \text{for } r > 0, -\pi < \theta < \pi$$

Prove that u is a harmonic function.

Problem 18 (Fa92). Let k be a positive integer. Determine those real numbers c for which every sequence (x_n) of real numbers satisfying the recurrence relation

$$\frac{1}{2}(x_{n+1} + x_{n-1}) = cx_n$$

has period k , i.e., $x_{n+k} = x_n$ for all n .